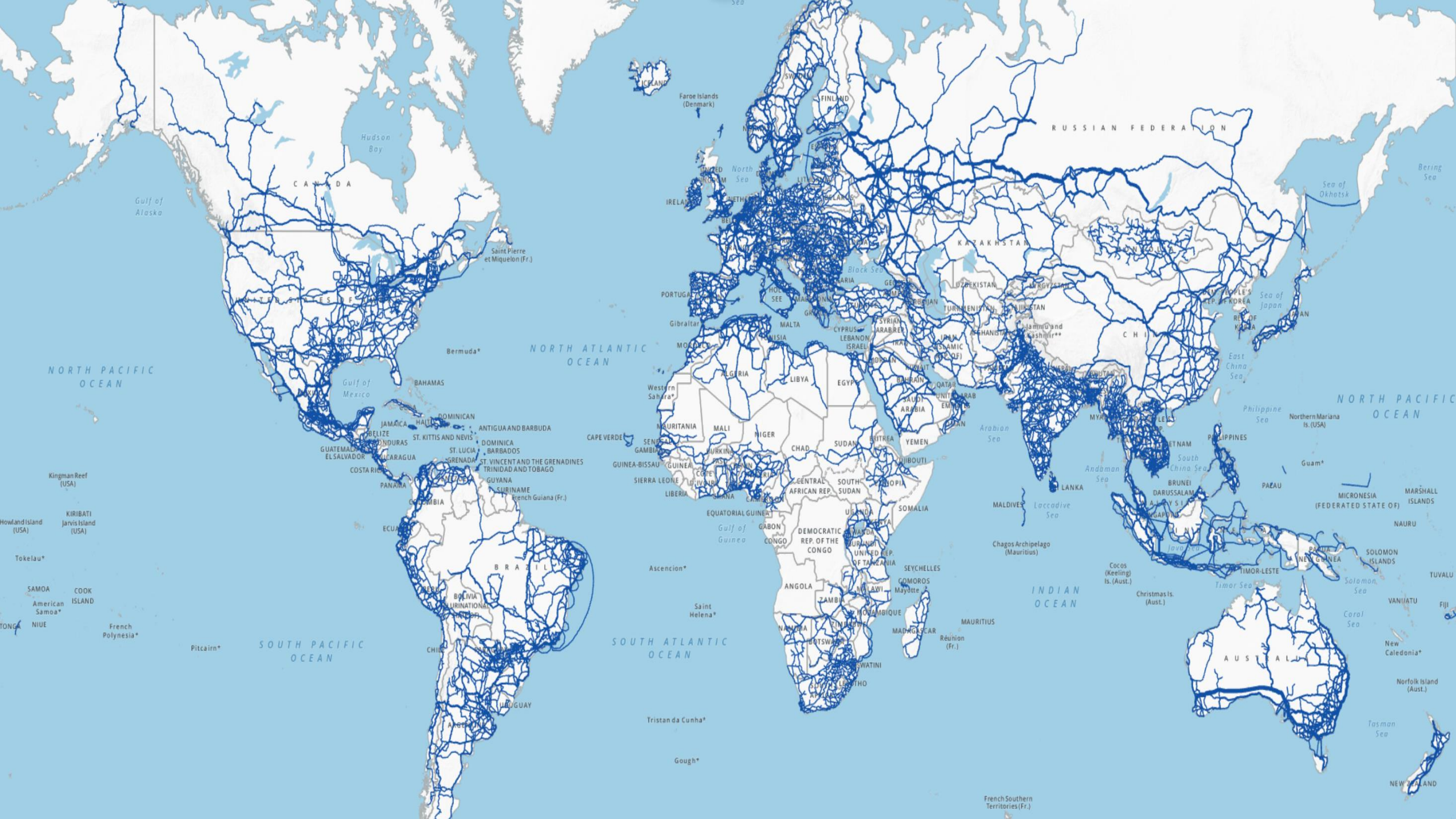


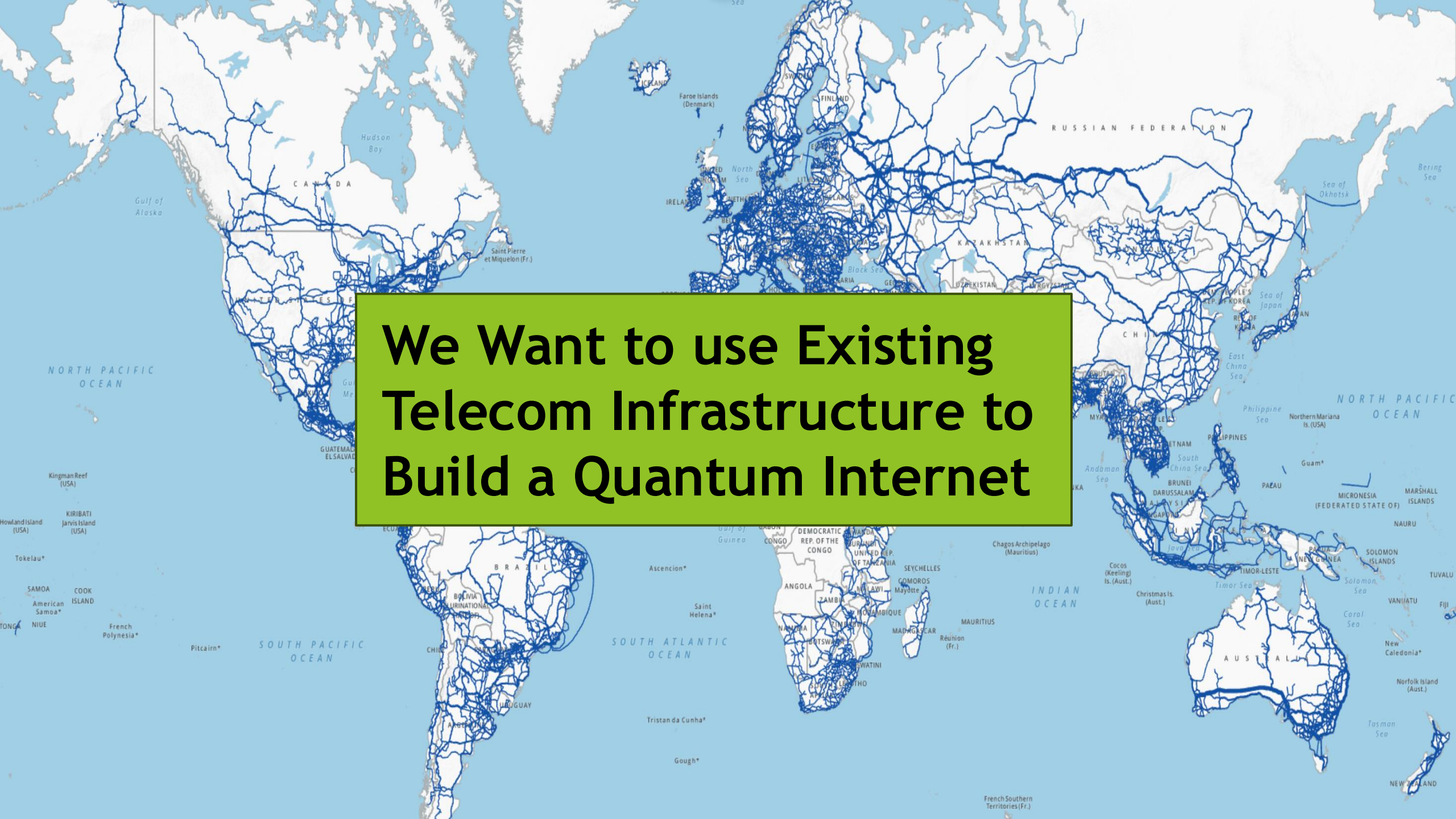
Simulation of Squeezed Light Generation in an Optical Resonator

Aaron Dayton

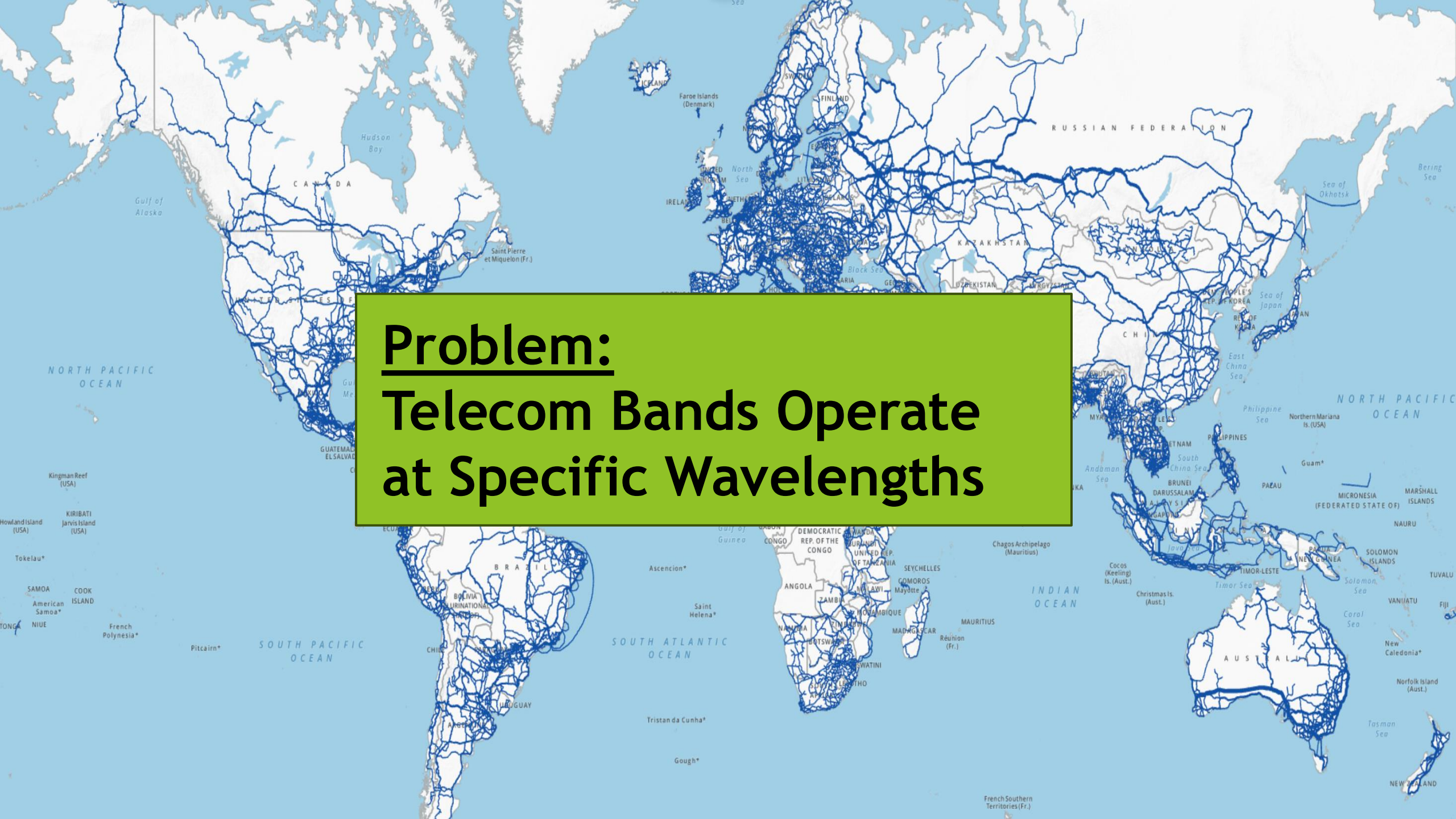
Supervisor: Andrew MacRae

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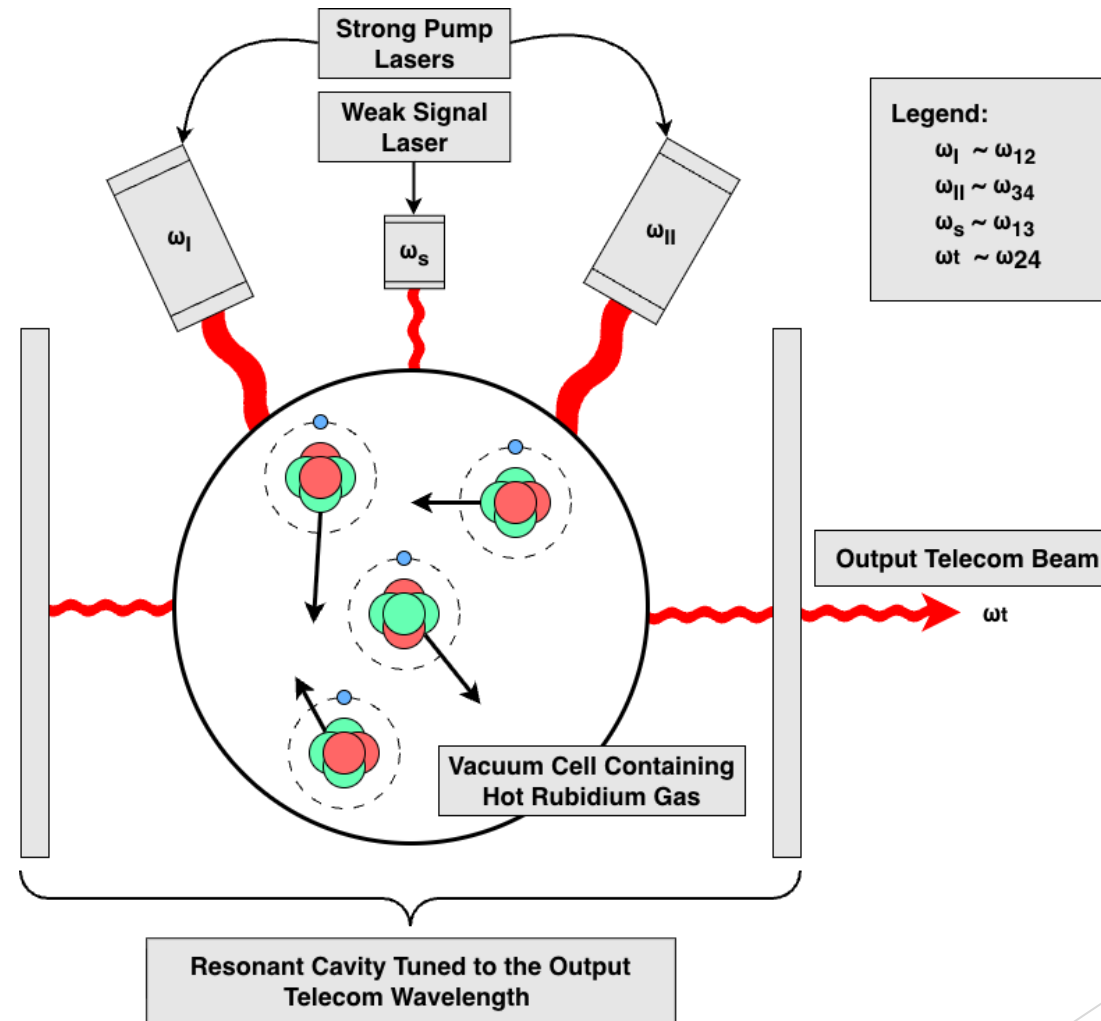
A world map with a dense network of blue lines representing telecom infrastructure. The lines are most concentrated in North America, Europe, and East Asia, with sparser coverage in South America, Africa, and Oceania. The map includes labels for major countries and bodies of water.

**We Want to use Existing
Telecom Infrastructure to
Build a Quantum Internet**

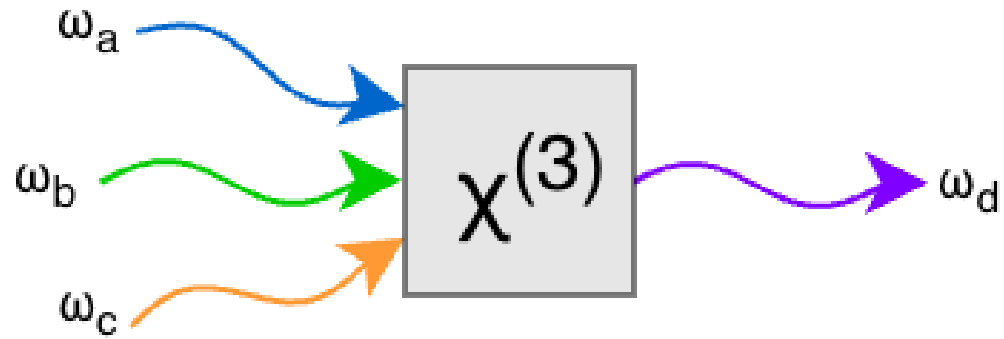
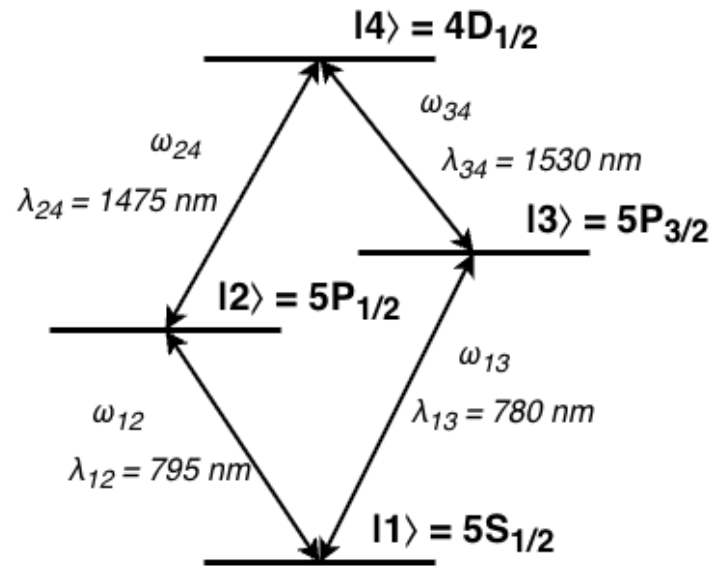
A world map with a dense network of blue lines representing telecom routes. The lines are most concentrated in North America, Europe, and East Asia, with a sparser network in South America, Africa, and Oceania. The map includes labels for major countries and regions, as well as various seas and oceans.

Problem: Telecom Bands Operate at Specific Wavelengths

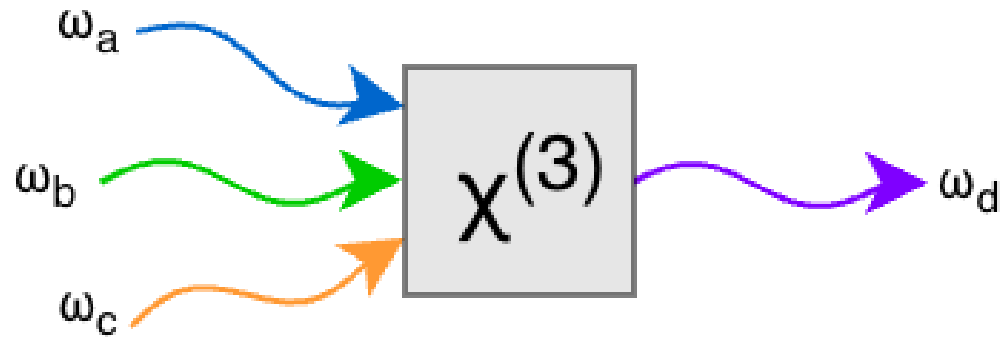
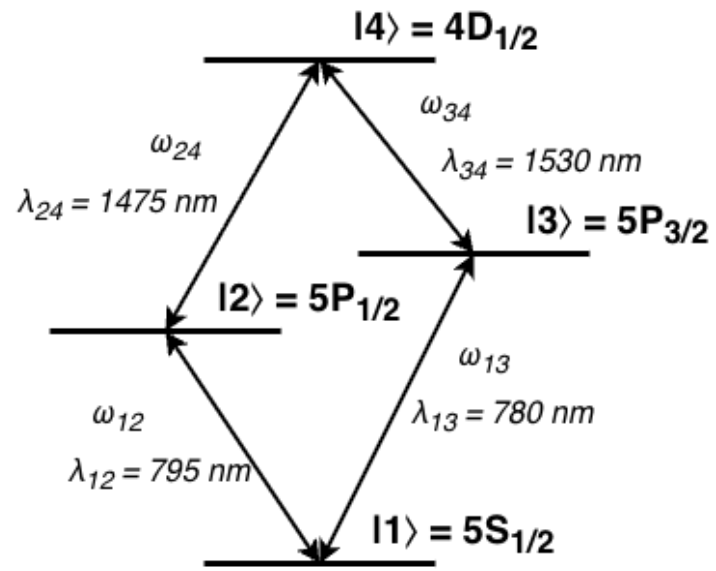
Frequency Converter Cartoon Setup



Four Wave Mixing



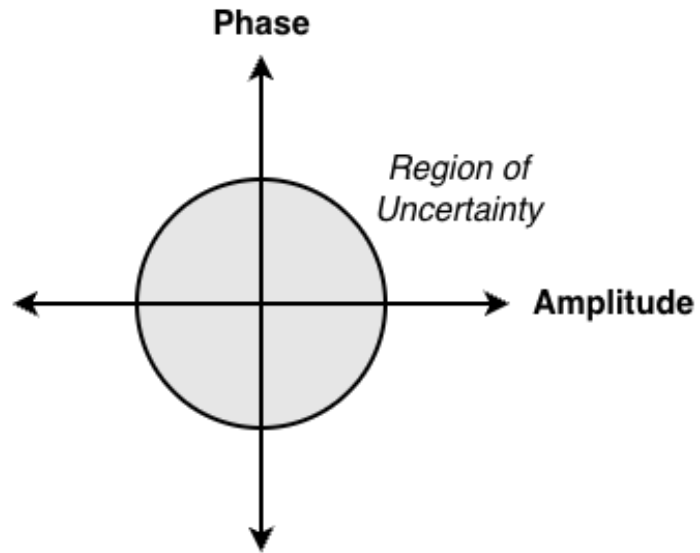
Four Wave Mixing



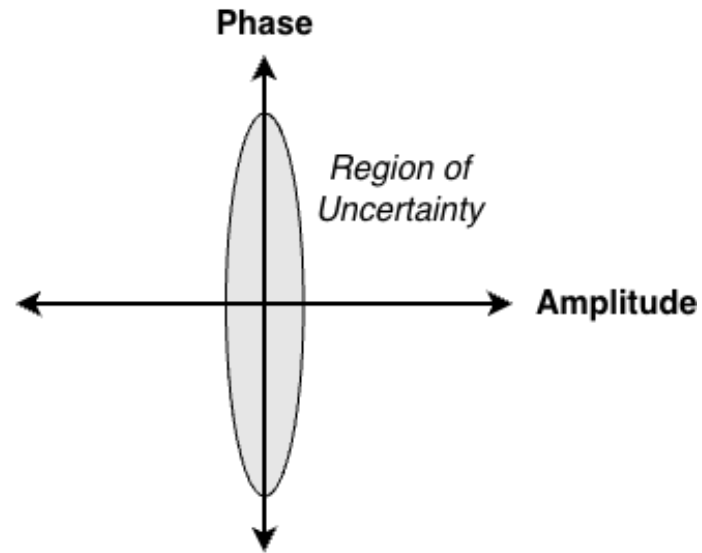
- Generate light in the telecom C-band (1530 nm to 1565 nm)

Four Wave Mixing Generates Squeezed Light

Normal Light Quadratures



Squeezed Light Quadratures



$$\Delta \hat{x} \Delta \hat{p} \geq \frac{\hbar}{2}$$

Electromagnetically Induced Transparency

$$E \propto e^{-\text{Im}(\chi^{(1)})kz/2}$$

Electromagnetically Induced Transparency

$$E \propto e^{-\text{Im}(\chi^{(1)})kz/2}$$

- Use EIT to minimize decay within medium

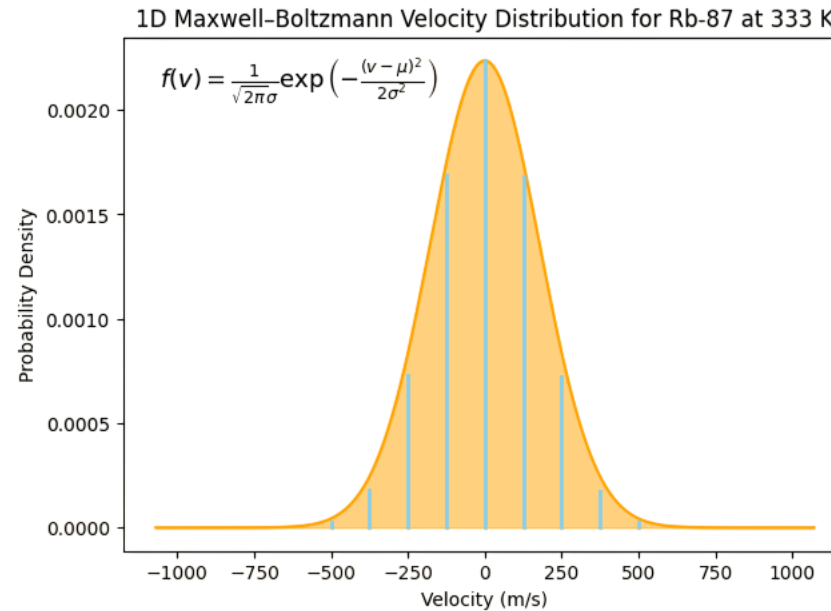
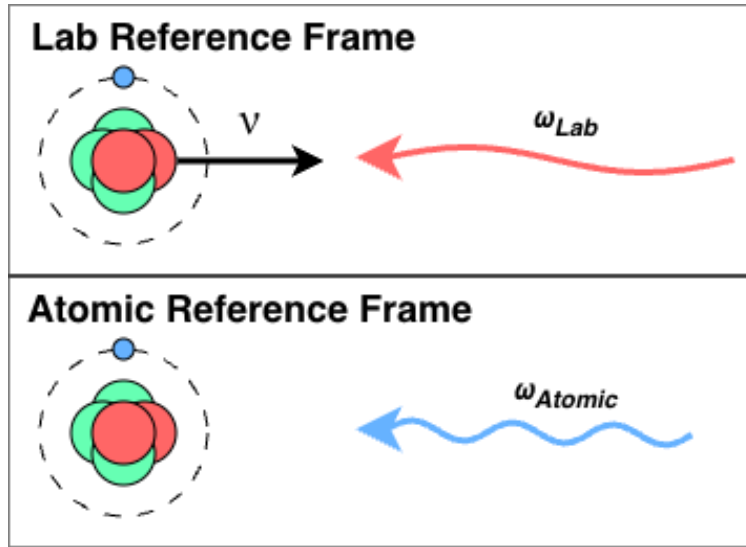
Electromagnetically Induced Transparency

$$E \propto e^{-\text{Im}(\chi^{(1)})kz/2}$$

- ▶ Use EIT to minimize decay within medium
- ▶ EIT for the output transmission beam is desirable

Problem:

Things move (Doppler Broadening)



Solution Strategy

- 1) Solve for steady state

$$\frac{d\hat{\rho}_S}{dt} = -i [\hat{H}_S, \hat{\rho}_S] + \sum_{\alpha} \gamma_{\alpha} \left(\hat{L}_{\alpha} \hat{\rho}_S \hat{L}_{\alpha}^{\dagger} - \frac{1}{2} \{ \hat{L}_{\alpha}^{\dagger} \hat{L}_{\alpha}, \hat{\rho}_S \} \right)$$

Solution Strategy

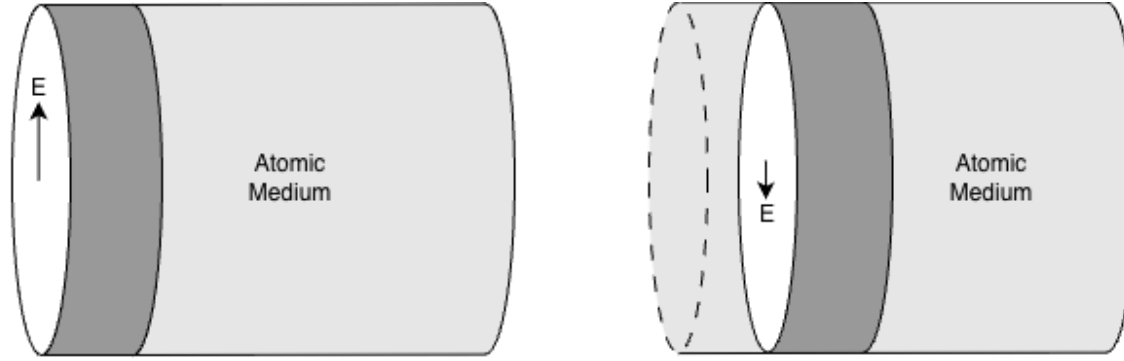
- 1) Solve for steady state

$$\frac{d\hat{\rho}_S}{dt} = -i [\hat{H}_S, \hat{\rho}_S] + \sum_{\alpha} \gamma_{\alpha} \left(\hat{L}_{\alpha} \hat{\rho}_S \hat{L}_{\alpha}^{\dagger} - \frac{1}{2} \{ \hat{L}_{\alpha}^{\dagger} \hat{L}_{\alpha}, \hat{\rho}_S \} \right)$$

- 2) Average over velocity distribution

$$\bar{\rho} \approx \sum_v w(v) \hat{\rho}^{(m)}(v)$$

Another Problem: Things Change (Maxwell-Bloch)



$$\frac{\partial}{\partial \zeta} \varepsilon(\zeta, \tau) = i \frac{\omega}{2n\epsilon_0 c} \pi(\zeta, \tau)$$

ε = Electric Envelope
 $|E| \propto \varepsilon$

Solution Strategy

- ▶ 1) Solve for steady state at one position

$$\frac{d\hat{\rho}_S}{dt} = -i [\hat{H}_S, \hat{\rho}_S] + \sum_{\alpha} \gamma_{\alpha} \left(\hat{L}_{\alpha} \hat{\rho}_S \hat{L}_{\alpha}^{\dagger} - \frac{1}{2} \{ \hat{L}_{\alpha}^{\dagger} \hat{L}_{\alpha}, \hat{\rho}_S \} \right)$$

- ▶ 2) Average over velocity distribution

$$\bar{\rho} \approx \sum_v w(v) \hat{\rho}^{(m)}(v)$$

- ▶ 3) Find the change in electric field strength

$$\frac{\partial}{\partial \zeta} \varepsilon(\zeta, \tau) = i \frac{\omega}{2n\epsilon_0 c} \pi(\zeta, \tau)$$

- ▶ 4) Repeat

Results

► System Hamiltonian

$$\tilde{H}_{ab} = \hbar \begin{pmatrix} 0 & \Omega_s \hat{a}^\dagger & \Omega_I & 0 \\ \Omega_s^* \hat{a} & \Delta_s & 0 & \Omega_{II} \\ \Omega_I^* & 0 & \Delta_I & \Omega_i \hat{b}^\dagger \\ 0 & \Omega_{II}^* & \Omega_i^* \hat{b} & \Delta_{II} + \Delta_s \end{pmatrix} + \hbar \Delta_a \hat{a}^\dagger \hat{a} + \hbar \Delta_b \hat{b}^\dagger \hat{b} + \hbar (\varepsilon_s \hat{a} + \varepsilon_s^* \hat{a}^\dagger) + \hbar (\varepsilon_i \hat{b} + \varepsilon_i^* \hat{b}^\dagger)$$

Results

► System Hamiltonian

$$\tilde{H}_{ab} = \hbar \begin{pmatrix} 0 & \Omega_s \hat{a}^\dagger & \Omega_I & 0 \\ \Omega_s^* \hat{a} & \Delta_s & 0 & \Omega_{II} \\ \Omega_I^* & 0 & \Delta_I & \Omega_i \hat{b}^\dagger \\ 0 & \Omega_{II}^* & \Omega_i^* \hat{b} & \Delta_{II} + \Delta_s \end{pmatrix} + \hbar \Delta_a \hat{a}^\dagger \hat{a} + \hbar \Delta_b \hat{b}^\dagger \hat{b} + \hbar (\epsilon_s \hat{a} + \epsilon_s^* \hat{a}^\dagger) + \hbar (\epsilon_i \hat{b} + \epsilon_i^* \hat{b}^\dagger)$$

► Maxwell-Bloch Equations

$$\partial_\zeta \Omega_s(\zeta, \tau) \approx i \frac{\omega_s \nu d_{12}^2}{2v_g \epsilon_0 \hbar} \bar{\rho}_{12}$$

$$\partial_\zeta \Omega_s(\zeta, \tau) \approx i \frac{\omega_i \nu d_{34}^2}{2v_g \epsilon_0 \hbar} \bar{\rho}_{34}$$

Developed Computational Tools

- ▶ 1) Solve for steady state at one position

$$\frac{d\hat{\rho}_S}{dt} = -i [\hat{H}_S, \hat{\rho}_S] + \sum_{\alpha} \gamma_{\alpha} \left(\hat{L}_{\alpha} \hat{\rho}_S \hat{L}_{\alpha}^{\dagger} - \frac{1}{2} \{ \hat{L}_{\alpha}^{\dagger} \hat{L}_{\alpha}, \hat{\rho}_S \} \right)$$

- ▶ 2) Average over velocity distribution

$$\bar{\rho} \approx \sum_v w(v) \hat{\rho}^{(m)}(v)$$

- ▶ 3) Find the change in electric field strength

$$\frac{\partial}{\partial \zeta} \varepsilon(\zeta, \tau) = i \frac{\omega}{2n\epsilon_0 c} \pi(\zeta, \tau)$$

- ▶ 4) Repeat

Thank You

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